Advanced Banking

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2025/26 - Period 2
https://studiegids.vu.nl/EN/courses/2025-2026/E_FIN_AB#/

Group assignment

Disclaimer: Views expressed do not necessarily coincide with those of DNB or the Eurosystem.
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Quick overview of group assignment

- Form groups via Canvas (max 4 students).
- Use two data sets already provided on Canvas
- Analyse link between net interest margin (NIM) and macro economy using regression model
- Run a stress test
- Submit video pitch + code file by Wed 26 November

Learning components

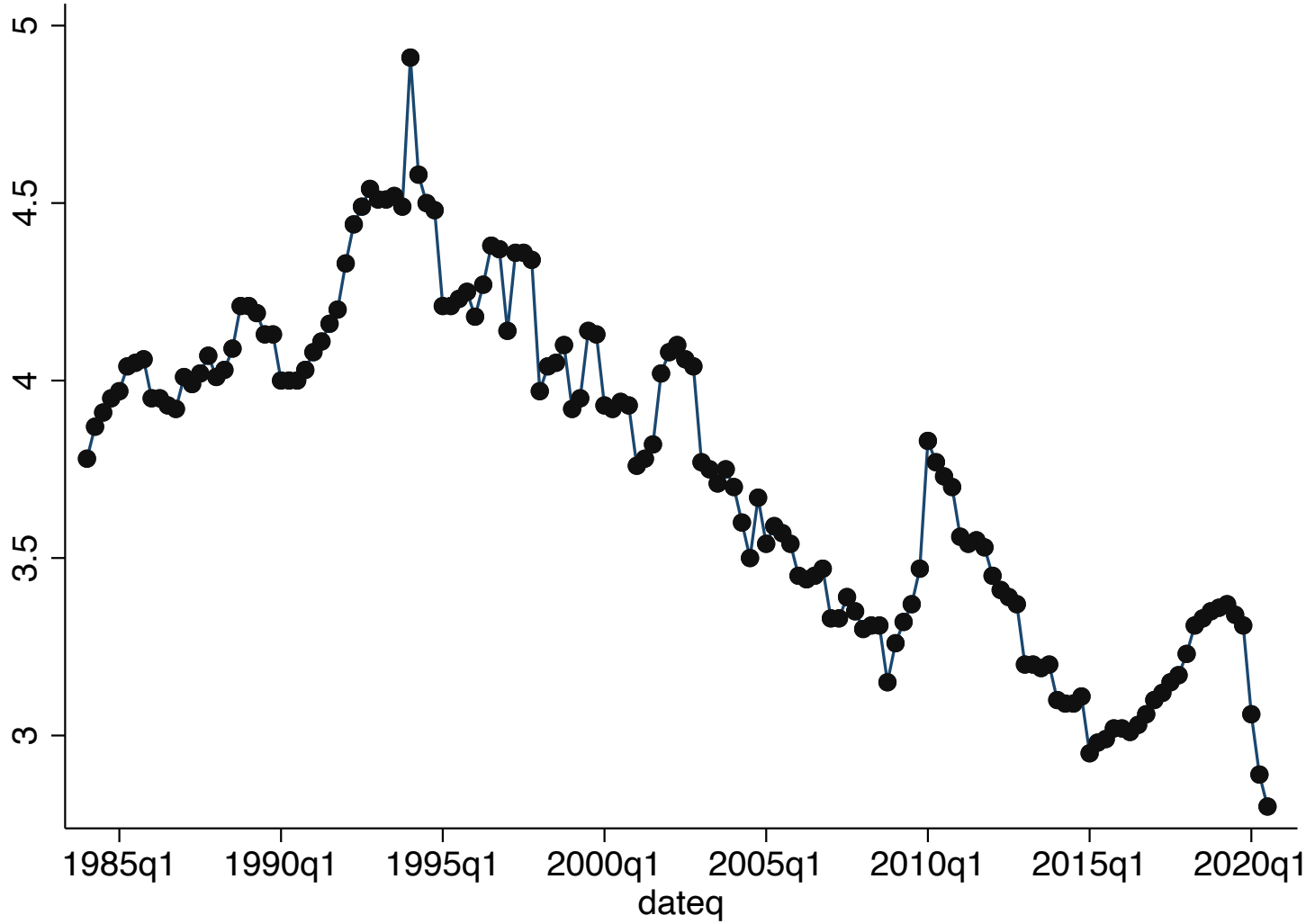
- **Academic skills**: data management, regression analysis, interpretation
- **Bridging theory and practice**: apply theoretical concepts to one bank
- **Professional / social skills**: working as a group, presenting results, documenting analysis

1. Background

Remember the bank's decision problem (stylized)?

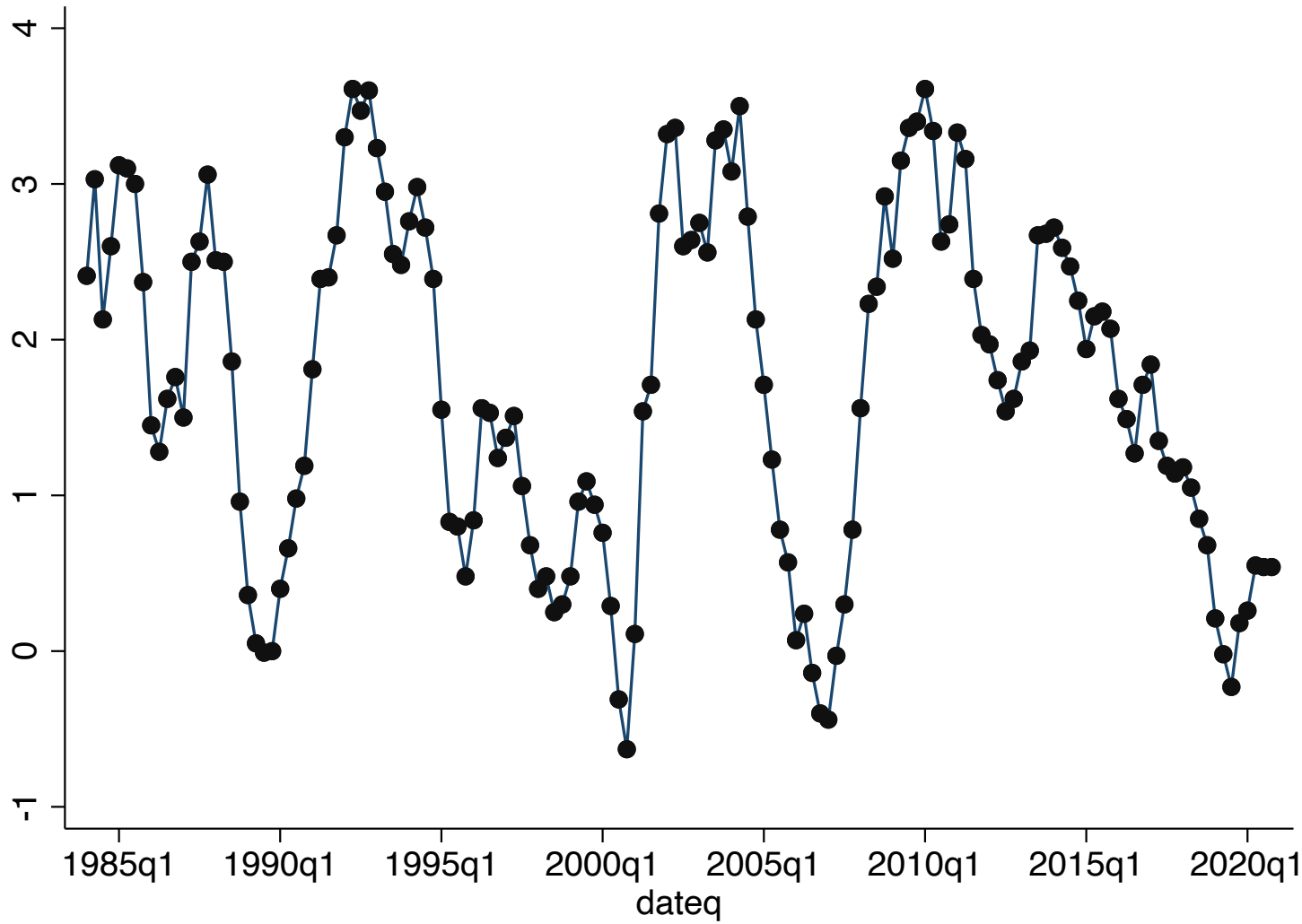
- Max $\pi = r_r R + r_l L + r_s S - r_d D - r_{of} OF$
- s.t. $R + L + S \leq D + OF + C$

Net interest margin of U.S. banks



Source: St Louis Fed.

Term spread in the U.S.



Source: St Louis Fed.

Return on assets of U.S. banks



Source: St Louis Fed.

Claessens et al. 2018

- Focus on effects of low interest rates (2010s)
- Panel regression for 3,385 banks in 47 countries
- Study net interest margin and return on assets
- One percentage point drop in interest rate implies 8 basis points lower NIM

2. Details on the assignment

What is the assignment about?

- Sign up to a group on Canvas.
- You will find two data sets on Canvas containing information on the banking sector and the macroeconomy in a hypothetical country (Z).
  - AB25_GroupAssigment_MacroData.xlsx : Year + 7 macrofinancial series for Z.
  - AB25_GroupAssigment_BankData.xlsx : Year / Bankid / NIM / Total assets
- Your group number on Canvas is the number of the bank (*Bankid*) you will focus on.

Questions to be answered

You are asked to place yourself in the role of a financial analyst who is tasked with monitoring this bank. Your manager asks the following:

- How concentrated is the banking system in Z?
- Based on a regression model, which macrofinancial factors determine the NIM of this bank?
- How does these factors compare to the literature (i.e. Claessens et al., 2018)
- What are strong and weak points of such a regression-based approach?
- How vulnerable would the bank's net interest margin be in case of a severe recession in Z during the next two calender years (i.e. a stress test for years 41 - 42)?

Deliverables

Your manager asks for two deliverables

- a presentation (i.e. the video pitch), containing main points
- some insight into your research approach (i.e., the code)

How to form groups?

- Students can form groups via Canvas.
- Group size: max 4

What about grading?

- See also the rubric on Canvas.
- Group assignment grade will be 15% of final grade for AB
- No resit possible.

What are the requirements for the video pitch?

- A maximum of 5 minutes
- Focus on key aspects: what is most important for the manager to know?
- Every member of the group needs to participate actively in the pitch

What are the requirements for the code?

- You are free to choose software package
- Make sure the code is clearly commented
- Ideally, submit everything in one file
  - preparing data / regression /scenarios

What are the timelines?

- Week 1: Explanation in tutorial
- Week 3: Opportunity to ask questions in tutorial
- Week 5: Submit video pitch + code file via Canvas
  - Deadline: Wednesday 26 November, 18:00

Best of luck with the group assignment!